The small port, I, is provided for the purpose of supplying a small quantity of steam to the pump, sufficient to keep the latter moving, while steam valve 5 is on its seat, thus providing a circulation that prevents condensation of steam in the steam pipe, and in cold weather prevents freezing up of the pump.

There are several combinations of the pump governor to suit particular kinds of service, such as the single, the duplex, and the triplex governor. These together with the defects to which the governors are occasionally subject, will be treated in another article.

J. P. KELLY.

Watertown, N. Y.

QUESTIONS AND ANSWERS

ON THE AIR BRAKE

(65) S. J. B., Trenton, N. J., writes: I have noted that on one of our high-speed trains coming to a stop there was a hissing noise of the triple valve which finally went into the well-known sound of the triple valve releasing the brake. I would ask what makes this peculiar release with the high-speed brake? A.—This hissing sound is probably the pressure reducing valve of the high-speed brake releasing brake cylinder pressure, due to a full service application, and while the reducing valve was still blowing the engineer released the brake, thus stopping the discharge at the reducing valve and creating a new one at the exhaust port of the triple, the sound with which we are all quite familiar.

(66) J. C. M., Packerton, Pa., asks: How do you tell the difference between a leak from the auxiliary reservoir into the brake cylinder and one from the train line to the auxiliary? A.—Train pipe pressure leaking into the brake cylinder, past the rubber seated emergency valve, the gasket between the lower case of the triple valve and the main body, or any other passageway, will be manifested by a blow at the exhaust port of the triple valve the instant pressure is turned into the train line. However, if the leak be one from the auxiliary reservoir into the brake cylinder, past a leaky slide valve, broken gasket between the auxiliary reservoir and brake cylinder, or defective tube in the auxiliary reservoir, this leak will be manifested by a very slight leakage as pressure is first turned into the auxiliary, but gradually increasing as the auxiliary reservoir charges up.

(67) P. S., Frankfort, Ind., writes: In the February issue of RAILWAY AND LOCOMOTIVE ENGINEERING I saw a cut of an automatic sight feed lubricator for the air cylinder of the air pump. Do you think they are a great enough advantage over the ordinary cup now in use to make them advisable? A.—The service of the air pump on long trains of to-day is much more severe than formerly with shorter trains. Doubtless the old form of cup was adequate and suited to that service, but not so well suited to the heavy service of to-day as is the regular feeding automatic cup. The sight feed feature doubtless aids the engineering considerably in advising him whether the cup is feeding, and when it has fed out. Undoubtedly, experience to date has proved that the automatic sight feed cup for the air cylinder of the air pump is a decided advantage.

(68) P. E. M., Murray Bridge, So. Australia, writes: What is the advantage of placing the release valve of rolling stock air brake on the auxiliary reservoir? Some vehicles have this valve on the brake cylinder. I fail to see any advantage in the placing of this valve on the auxiliaries. By what I can see of it, it seems to me that the auxiliary attachment is rather to a disadvantage. A.—With the earlier forms of air brakes, the release valve was placed on the brake cylinder, but on account of it being left open so frequently after bleeding off the brake, thereby causing trouble and even expensive accidents, because its open condition was never known until the brake was applied, the release valve was removed and placed in the auxiliary reservoir. Now, when it is desired to release a brake with the release valve, the triple valve is made to move to release position, and the brake releases in the natural way, which is much quicker than if the release was made with the brake cylinder, where both the brake cylinder air and the auxiliary reservoir air sometimes have to be reduced to release the brake.

(69) B. C. G., New York city, writes: There are some complaints from one or two of the motormen of the Manhattan Elevated road saying that some sort of mufflers had been put in the exhaust ports of the triple valves, which results in making the air escape from the brake cylinder so slowly that it sometimes takes the control of the train away from the motorman. A.—The muffler placed in the exhaust port referred to is an ordinary pipe plug with a hole about 3-16 of an inch in size through it. This plug not only muffles the sharp whistling sound of the triple valve, but retards the discharge of pressure from the brake cylinder. It has been found necessary to retard and lengthen the exhaust because of the very short piston travel on these cars, the travel being between three and four inches. With this travel the air escapes so suddenly that it is very difficult to make a smooth stop. The lengthened exhaust acts more like a longer piston travel, and permits the train to be brought to a standstill with greater accuracy, at the same time avoiding the disagreeable shock experienced with the short travel and wide-open exhaust.

(70) J. R. McE., Little Rock, Ark., asks: Why does the slide valve feed valve attachment on some brake valves allow the pressure to creep up above 70 pounds what it is set for, and equalizes at 90 what the main reservoir pressure is set for? A.—The diaphram valve may be held from its seat by foreign substance or the slide valve or its seat be cut, thus permitting main reservoir pressure to leak into the train pipe. Again, the lead gasket between the brake valve body and the feed valve attachment may leak, causing the same trouble. Sometimes, in truing up of the slide valve seat, no heed is taken that the seat be kept parallel with the floor and roof of the piston's cylinder. The slide valve, fitting snugly between the shoulders of the piston stem, tends to slightly lift at one end or the other, thus tipping it and permitting main reservoir pressure to feed slowly under it into the train pipe. Sometimes, in screwing home the cylinder cap, the spring is twisted, and transmits the twist to the piston, which slightly tips the side of the slide valve from its seat, permitting main reservoir pressure to leak under the edge into the train pipe.

(71) P. E. M., Murray Bridge, So. Australia, writes: I start the pump and accumulate 80 pounds in the main reservoir, and having handle of driver's valve in release and full-charge position, I understand that both pipes (train and reservoir) contain an equilibrium of pressure. Now I put the handle in running position and a decrease of 8 to 10 pounds train line pressure is shown by the gauge pointer. Why is it? Our engines are fitted with the old-type driver's valve with 10-pound reserve valve, and all plain triples. A.—If the train line pressure drops when you bring your brake valve handle from release to running position, there is a leakage of pressure from the train pipe at some point. If the main reservoir pointer goes up, and the train line pointer stands still, there is no leakage in the train pipe, and the difference between the two hands indicates merely a rise of pressure in the main reservoir. If, after coupling the engine to the train, with the handle in full release position, the black pointer drops and continues to drop when the handle is brought to running position, it shows that pressure is passing into the train pipe in full release position faster than the feed groove in the triple valves can take the pressure from the train line. When the handle is returned to running position, and the black pointer falls again, it shows that the feed grooves at the triple valves are taking train line pressure faster than it can be supplied by the running position of the brake valve.

QUESTIONS AND ANSWERS

On Air Brake Subjects.

(16) R. E. J., Pueblo, Colo., asks:

Is the 3-position retaining valve now being issued as part of the standard equipment, or is it only out on trial? A.—It is a regularly listed part of the Westinghouse standard equipment.

(17) J. L. E., Battle Creek, Mich., writes:

I hear that F-46 triple valve is to be supplied in the future for all locomotives with brake cylinders more than 10 inches in diameter, instead of the special engine triple; how about it? A.—Your understanding is correct.

(18) C. P. McG., Buffalo, N. Y., asks:

What are the inside dimensions of the Westinghouse cast iron auxiliary reservoir for the 8-inch freight brake cylinder? I wish to find the cubic contents or volume. Also, what do they call the cubic contents of the 10x33 auxiliary reservoir used with the 10-inch brake cylinder? A.—The cubic capacity of the cast iron auxiliary reservoir for the 8-inch freight brake cylinder is 1,620 cubic inches. The capacity of the 12x33-inch auxiliary reservoir for 10-inch brake cylinder is 2,156 cubic inches.

(19) N. R. E., Buffalo, N. Y., writes:

1. How can you distinguish between the three kinds of Westinghouse plain triple valves, the G24, the F25, and the F46? A.—1. The "name" of each triple is cast on its body. 2. What is the difference in these triple valves? A. 2—There is a slight difference in size of the ports and parts to adapt each to its special work. The G24 triple is for 8 and 10-inch tender brake cylinders on freight and switching engines, the H24 for 8 and 10-inch driver brake cylinders on all kinds of engines, the F25 for 12-inch tender-brake cylinders on freight and switching engines, and the F46, or "high speed" triple valve, for all 12, 14 and 16-inch driver brake cylinders.

(20) B. B. C., Chicago, Ill., asks:

Is the omission of the train pipe from the cuts of driver and engine truck brakes in Figs. 4 and 6, on page 27 of January number a mistake? A.—Not a mistake, but an oversight. They might as well have been put in; but as the main object was to show the location of the cut-out cocks in the auxiliary reservoir pipes and brake cylinder pipes, and as the location of the triple valve would do this without the aid of the main train pipe, the omission of the train pipe, although accidental, was not considered serious.

(21) B. B. C., Scranton., Pa., asks:

Why is a larger brake valve reservoir being supplied? What are its outside dimensions? In what service is it used? A.—The Westinghouse Air Brake Co has decided to make a change in the size of their standard equalizing reservoir for brake valves. As you know, the present reservoir is 10x12 inches. The modified reservoir will be 10x14 1-2 inches. This change was brought about by the desire of the Pennsylvania R. R. to increase the size of the equalizing reservoir, to prevent a possible quick-action application of the brakes when making a service application on a high-speed train, when the train contained but a few cars. Experience in the past on other roads has not suggested the advisability of such a change, but it has been thought best to eliminate all possibility of any tendency to get quick-action application with the high speed brake in making a service application.

(22) J. B. S., Hamilton, New South Wales, writes:

I was working an engine the other day, and when working with light engine only, in making a service application, the tender brake would go on properly, and remain on; but when connected to a train, it would not stay on five seconds. It would just go on and come off again every time. A.—The trouble is probably due to leakage in both the train line and auxiliary reservoir. With the light engine these two leakages will balance each other, and the brake will remain set. When the train is coupled on, the train-pipe volume is greater, and if the train line of the cars does not leak, the brake on the tender will bleed itself off by auxiliary reservoir leakage. Again, the triple piston packing ring may be in a sufficiently bad condition to permit the train pipe pressure on the light engine to leak past the ring into the auxiliary reservoir, keeping the pressures balanced, regardless of the auxiliary reservoir leakage. When the engine is coupled to a train, the train pipe volume would be increased to a degree that would not permit the train line and auxiliary reservoir pressures to equalize, and the auxiliary reservoir leakage would bleed the tender brake off.

(23) B. M. McC., Louisville, Ky., asks:

If you gradually draw all of the air out of a train line, will the graduating spring compress and will the valve move into emergency position? A.—Yes. To illustrate: Suppose we charge the train line and auxiliary reservoir to 70 pounds pressure. Now draw off 20 pounds from the train line, setting the brake full. We will now have 50 pounds pressure in the brake cylinder, auxiliary reservoir and train pipe each. Now, again, suppose we reduce the train pipe pressure 10 pounds more, thus making the train line pressure 40 pounds. As there has been no outlet for pressure from the auxiliary reservoir and brake cylinder, the pressures will remain the same therein. Consequently, the 50 pounds in the auxiliary reservoir being greater than the 40 pounds in the train line, and the resistance of the graduating spring, the triple piston and slide valve will travel to their furthermost limit, compressing the graduating spring. Of course, the graduating spring would also be compressed if all the pressure were drawn from the train line.

(24) J. B. S., Hamilton, New South Wales, writes:

On one or two of our engines here (all triples are set in slow action in this state) the tender brake would not operate, and by cutting the triple into quick action position, the air brake would go on with a "bang," even though the brake valve handle was placed in full release position, and the brake cylinder "bled." The only way it could be released was by placing the triple three-way cock in slow active position again, then the brake would come off immediately. Leave the brake valve handle in full release, and by simply pulling the three-way cock down to quick action position, the brake, as above stated, would go on with a regular "bang" every time. A.—There is something wrong with the quick action features of the triple valve. The emergency valve is defective, or some foreign matter has lodged on its seat, holding the valve open. In this event, and with three-way handle in quick action position, the train-pipe pressure would have a free and constant passageway to the brake cylinder and would remain on as long as there is pressure in the train pipe. Possibly the leather packing in the brake cylinder is faulty. If this be true, a slow admission of pressure to the cylinder, such as service application would give, will leak out quickly and perhaps as rapidly as it enters. When the triple is cut into quick action position, the train-pipe pressure would go into the brake cylinder faster than it could leak out. Clean the triple valve parts thoroughly, paying particular attention to the quick action mechanism, and note that the brake cylinder packing is in good condition.

(25) B. J. K., St. Paul, Minn., writes:

I do not understand from the sketch in February RAILWAY AND LOCOMOTIVE ENGINEERING how I am to drill the hole in the brake valve, or to strike the port for the governor to give the high pressure when the brake is applied, and the low pressure when the brake is off. A—Draw a center line through the rotary valve seat and the part on the side of the brake valve to which the feed valve attachment is fastened. Draw a parallel line three-quarters of an inch to the side of this line—to that side nearer the main reservoir connection. Set your dividers for 2 67-128 inches, and from the center of the rotary valve seat scribe a circle through the parallel line. The point where this circular line intercepts the parallel line should be the center of your hole to tap into the port to the brake valve.